GANPAT S. SAWALE

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ABOUT ME

Electronics & Communication Engineering graduate with hands-on internship experience in embedded systems and IoT development at Cojag Technology Pvt. Ltd. Skilled in hardware interfacing, circuit design, and microcontroller programming using Arduino, ESP32, and Embedded C. Seeking a fresher Embedded Systems Engineer role where I can apply practical industry exposure to build reliable, efficient embedded solutions.

EDUCATION

Bachelor of Technology – Electronics & Communication Engineering (2022 – 2026)

Agnihotri College of Engineering, Wardha (RTMNU) | Year of Passing: 2026

Higher Secondary Certificate (HSC)

Madhomati Junior College, Lakh – Maharashtra State Board | Percentage: 72.83% | Year of Passing: 2022

INTERNSHIP EXPERIENCE

Intern – IoT

Cojag Technology Pvt. Ltd., Nagpur | Jan 2025 – dec 2025

- Interfaced and tested temperature/humidity and more sensors with Arduino as part of an IoT automation system; identified and resolved a hardware wiring fault during testing, restoring correct sensor communication.
- Developed and validated Embedded C programs for microcontroller-based prototypes, gaining hands-on exposure to industry debugging and hardware integration practices.

PROJECTS

Smart Cooler | Arduino UNO, DHT11, Embedded C

Automated cooling system that activates above 30°C with water level monitoring via LED indicators and buzzer alerts. Core automation functional; threshold calibration ongoing.

Offline AI-Based Home Automation System | Arduino Nano, AI Thinker VC02, Relay Module

Built an offline voice-controlled home automation system using AI Thinker VC02 for local voice recognition and Arduino Nano for relay-based appliance control via UART. Achieved voice recognition accuracy system is fully operational for light, fan, socket, and oven control.

SKILLS

Programming:	Embedded C, C,C++
Microcontrollers:	Arduino UNO, ESP32, NodeMCU, ATmega32, PIC, 8051
Communication Protocols:	UART, USART, SPI, I2C
Hardware:	Sensor Interfacing, Circuit Design, Hardware Debugging
Tools:	KiCad, EasyEDA,Proteus, Keil uVision5, MAPLAB X IDE, Microchip Studio

CERTIFICATIONS

Advanced Embedded Systems

SOFCON India Pvt. Ltd., Pune | NSDC & Skill India Certified

modules: Basic Electronics, Embedded C/ C Programming, Arduino UNO ,8051, AVR (ATmega32), PIC (PIC16F877A), and IoT (ESP32). PCB Design